

Light Flicker Detection using High-Speed Imaging

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Abstract — Light flicker (LF) phenomenon represented a concern for the researchers for several decades and is still considered one of the most important power quality problems due to its ability to be directly perceived by human beings causing significant physiological discomfort, physical and psychological tiredness, and even pathological effects. More than that, light flicker can trigger headaches, migraines, fatigue, epilepsy, and other neurological responses. The aim of this paper is to present a method for collecting the waveforms corresponding to the luminous flux modulation for different light sources using a high speed camera. The light flicker presence detection as well as its frequency are characterized based on the obtained waveforms plots using FFT techniques.

Keywords — light flicker, flicker detection, signal processing, power quality

I. INTRODUCTION

Flicker is defined in classical literature as an amplitude modulation of the voltage which has a modulation frequency of 0,5 Hz to 35 Hz. Arc furnaces, welding machines, generators like wind turbines and variable frequency drives are accepted as voltage flicker generating sources. The effect of the each source to the flicker contribution depends on the frequency and the magnitude of the voltage fluctuation. The frequency and the magnitude of flicker can be defined by a flicker meter, [1].

Still, the upper limit for the light variation frequency of 35 Hz imposed by the above mentioned definition is too low when the light flicker is considered, taking into account different studies that showed that flicker can be directly observed and perceived by human eyes even for frequencies of 70 up to 90 Hz, [2], [3] even if the greatest likelihood of seizures is for frequencies in the range of 12 Hz to 30 Hz as can be observed in Fig. 1, [4], [5]. It shows the distribution of the percent of people with a photoparoxysmal EEG response to intermittent photic stimulation at different flash rates.

Even if most humans are unable to perceive flicker in light above 60-90 Hz, there still remain measurable biological effects above the critical fusion frequency (CFF). Electroretinogram analyses responses indicate that invisible flicker is transmitted through the retina, even up to 200 Hz and may lead to headaches and eye strain [3], [6]. CFF is specific to each individual and refers to the frequency at which flicker is no longer perceived when a flickering source is observed directly, and the estimation of flicker fusion frequency usually

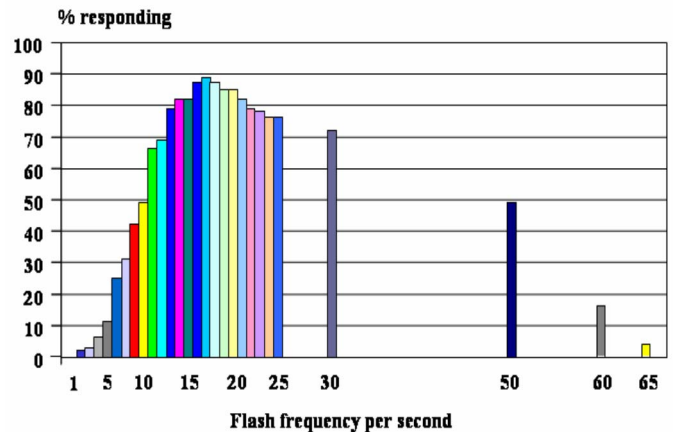


Fig. 1. Histogram depicting the percentage of known photosensitive subjects who show EEG photosensitive responses at particular flash frequencies [3], [4].

take no account of eye movement. It represents the limit between the visible and invisible flicker.

Considering the reasons presented above, the light flicker can be more generally defined as a rapid and repeated change over time in the brightness of light, [3]. It is also defined in the standard IEC 61000-4-30 as impression of unsteadiness of visual sensation induced by a light stimulus whose luminance or spectral distribution fluctuates with time [7].

Light flicker exists in most light sources, but with varying levels. This is true for the different types of lighting technologies such as incandescent, fluorescent, and solid-state lighting (SSL). In particular, SSL flicker is dependent on the method that is used to convert the input electric signal from a typical ac input to the desired dc output that the LEDs utilize [2]. Similar light flicker unwanted effects can be also produced by other light sources types such as the PC and TV, referring especially to the content of the software and to the programs corresponding to the unhealthy light emissions [8].

II. LIGHT FLICKER MEASUREMENT

For evaluating the light flicker there are two measures commonly used, according to Illuminating Engineering Society, as mentioned in [3], proposed by lighting designers: *Flicker Index* and *Percent Flicker*. The first measure is usually considered for determining the cyclic variation of the light output for different light sources. Considering the plot presented in Fig. 2, *Flicker Index* can be defined as the ratio

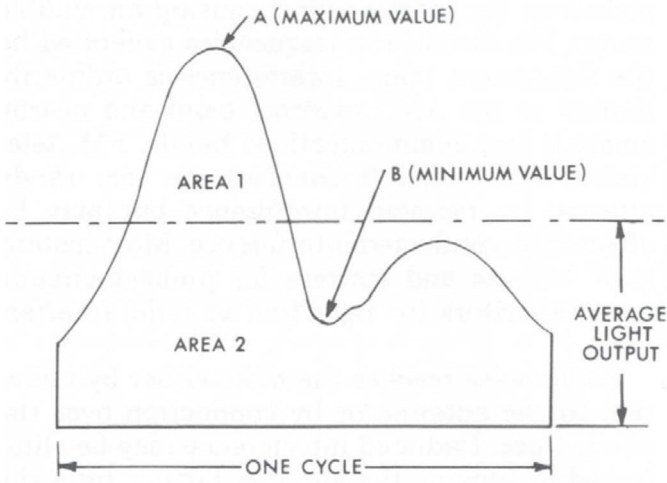


Fig. 2. Diagram for definition of flicker index and percent flicker, [3].

between the area above the average light output and the total area defined by the light output for a single cycle, as presented in (1). *Percent flicker* is mathematically described by (2):

$$\text{Flicker Index} = (\text{Area 1}) / (\text{Area 1} + \text{Area 2}) \quad (1)$$

$$\text{Percent Flicker} = 100 (A - B) / (A + B) \quad (2)$$

According to [3] the *Flicker Index* is preferred over *Percent Flicker* that indicates the peak-to-peak contrast. However, *Percent Flicker* is more often taken in consideration in research fields such the visual science or photobiology.

III. EXPERIMENTAL SET-UP

The method proposed in this paper for light flicker detection is based on collecting the waveforms corresponding to the luminous flux modulation using a high speed camera. The light source images are captured using a monochrome Fastec Imaging InLine camera with a frequency of 1000 frames/second and 400 μ s exposure, the images being stored in the camera internal memory and after transferred to a PC using an ethernet connection, as shown in Fig. 3. Typical images obtained for 15 W SSL lamp and a 30 W magnetically ballasted fluorescent lamp are presented in Fig. 4.

The images are afterward processed using a custom MATLAB program developed by the authors. Each image is decomposed into three matrices having the dimensions equal with the image resolution (480x260 pixels) according to RGB (Red, Green and Blue) color system. Each cell contains an integer number from 0 to 255 corresponding to the intensity of each of the three colors. For each of the three matrices the average of the cells is computed in order to determine the average intensity of color. The average color corresponding to the processed image is afterwards obtained using (3):

$$\text{RGB} = R \cdot 65536 + G \cdot 256 + B \quad (3)$$

A combination of the three components of RGB = (255, 255, 255) will provide the white color, RGB = (0, 0, 0) the

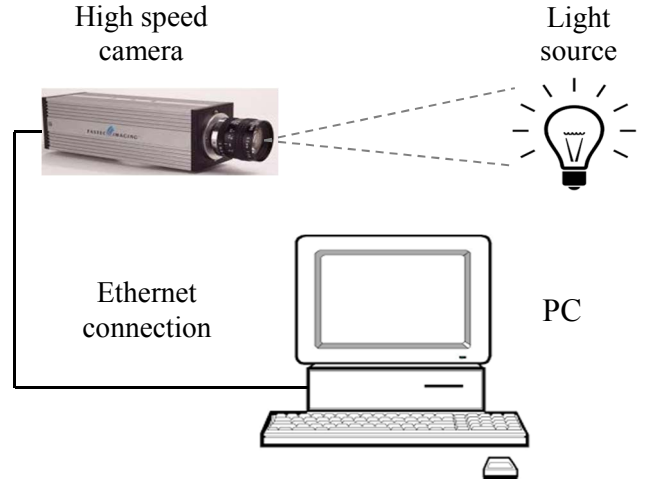


Fig. 3. Experimental set-up.

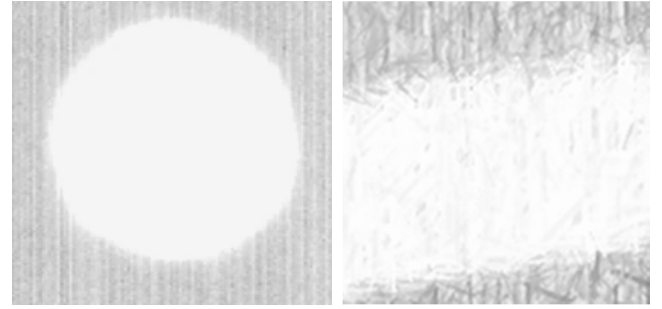


Fig. 4. Typical images recorded for a 15 W SSL lamp on the left and for a 30 W fluorescent lamp on the right.

black color and each combination RGB = (x, y, z), where $x = y = z$, will describe a gray color. As high the values of x, y and z as whiter the image is.

Using the set-up and the program presented above we can use the camera optical sensor as a photosensor. The resulting x, y and z values are higher if the intensity of the light given by the light source is higher. In order to minimize the measurement errors, the distance between the camera and the light source should be carefully chosen in order to avoid the camera saturation phenomenon.

Processing all the images recorded for a certain period of time, normalizing the obtained values and integrating them in time, we can create a plot corresponding to the light intensity variation in time.

IV. RESULTS AND DISCUSSIONS

For this study several types of light sources have been considered: 30 W magnetically ballasted fluorescent lamp, 6 and 15 W SSL A-lamp, 75 and 100 W incandescent bulbs. We have also analyzed the light emitted by a PC monitor while running a flicker generating program Power Quality Teaching Toy 3.0.4 designed by Alex McEachern, [9].

Fig. 5 shows light intensity variation measured for a magnetically ballasted fluorescent lamp using the proposed method. The obtained plot is similar to the ones presented in

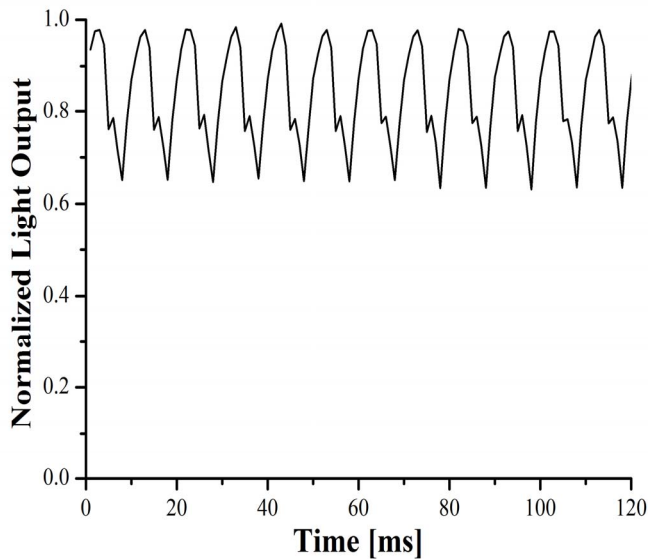


Fig. 5. Experimental data of flicker for a 30 W fluorescent lamp.

literature, such in [2] and [3]. On the descending plot, a discontinuity in the light intensity variation can be observed, which is most likely related to the lamp ageing.

As it can be observed, the main flicker frequency is 100 Hz, corresponding to the double of 50 Hz supplying voltage. The given waveform can be easily used for flicker index and percent flicker estimation. The values obtained are 0.06 for the flicker index and 0.21 for the percent flicker. Although the values are in the same range with the ones presented in literature, they cannot be taken into consideration as the real values because we did not performed a system light intensity calibration.

By applying the FFT decomposition of waveforms (see Fig. 6) we can observe that next to the continuous flux there are also important light intensity variations (flicker) having frequencies of 100 Hz (double of supplying voltage frequency), 200 Hz, 300 Hz, 400 Hz as well as components with frequencies of 50, 150, 250, 350 and 450 Hz. There also exist small components corresponding to 0.33 Hz frequency. Using the FFT decomposition there can be easily and fast determined the light flicker frequencies. Also, the flicker intensity corresponding to different frequencies can be afterward estimated.

In the case of the 6 and 15 W SSL using light emitting diodes (LEDs) lamps the light intensity was constant during the tests, the flicker index as well as percent flicker being null in this case.

For the incandescent lamps the light intensity variation corresponds to a sine wave having a frequency of 100 Hz obtained from the double of the supplying voltage frequency, as can be observed in Fig. 7. In this case, for recording the images we have used a protective opaque screen that allows the light passing through in order to avoid the camera saturation in the regions corresponding to the bulb filament. Because the light intensity was dimmed by the screen we could not determine the percent flicker or the flicker index in this case.

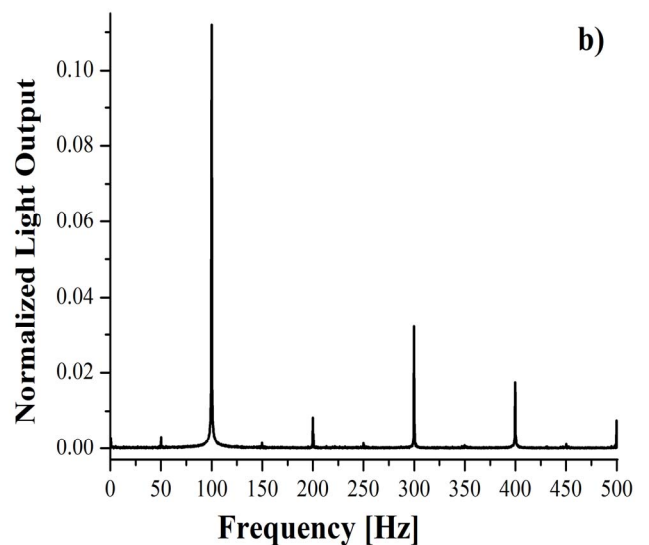
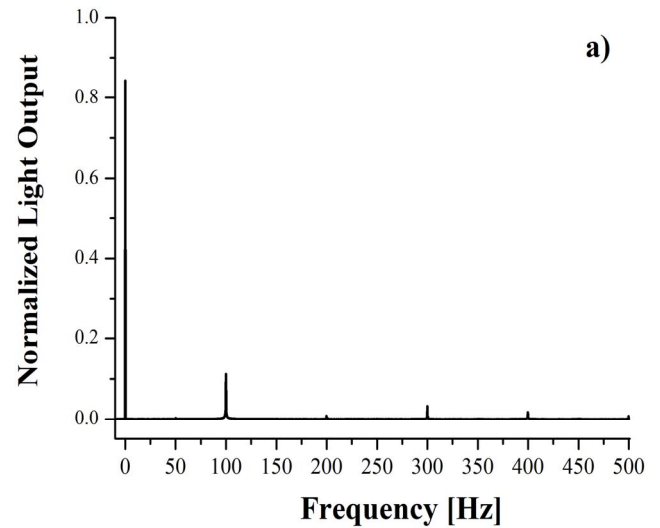


Fig. 6. Light spectrum obtained from FFT decomposition: a) full spectrum; b) DC component is omitted.

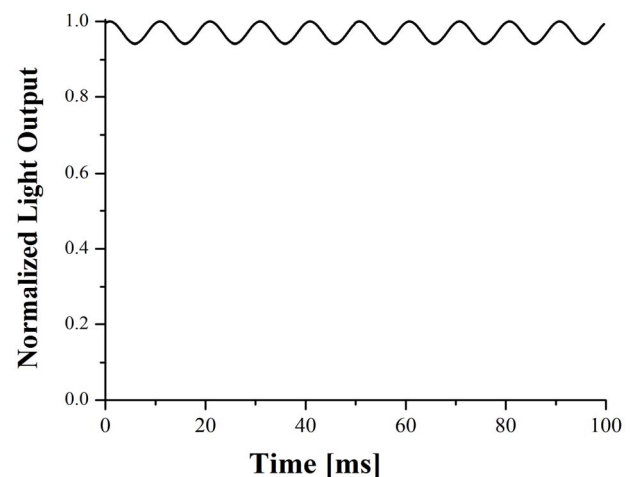


Fig. 7. Experimental data of flicker for a 75 W incandescent lamp.

In Fig. 8 and Fig. 9 are presented the light intensity waveform and the light spectrum obtained from FFT decomposition, respectively, in the case of a PC monitor while running the Power Quality Teaching Toy 3.0.4 flicker generating program. The program generates light flashes having different frequencies and amplitudes. For this experiment we have chosen the maximum amplitude and an intermediate value for frequency.

The image corresponding to the full PC monitor size was considered for being processed. The main flicker frequency indicated by the light intensity waveform FFT decomposition is 7.7 Hz, corresponding to the flashes repetition rate. A secondary variation having a frequency of 15.4 Hz can also be observed among small components with frequencies between 0 and 20 Hz.

In the case of the light sources represented by large diagonal PC monitors or TVs the utilization of this method has the main advantage that it allows recording the full size image

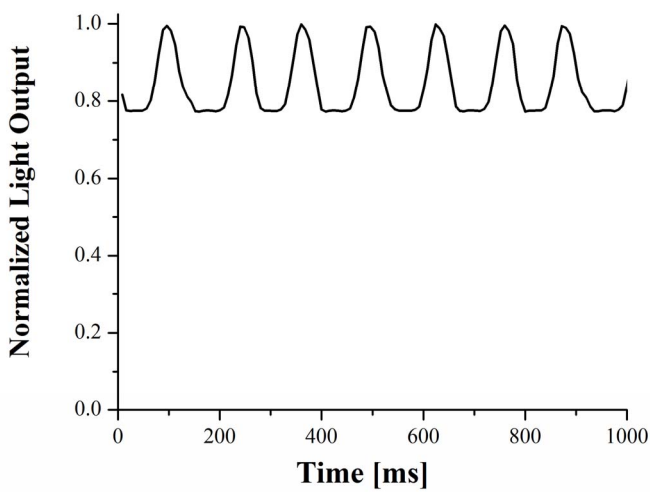


Fig. 8. Experimental data of flicker for a PC monitor.

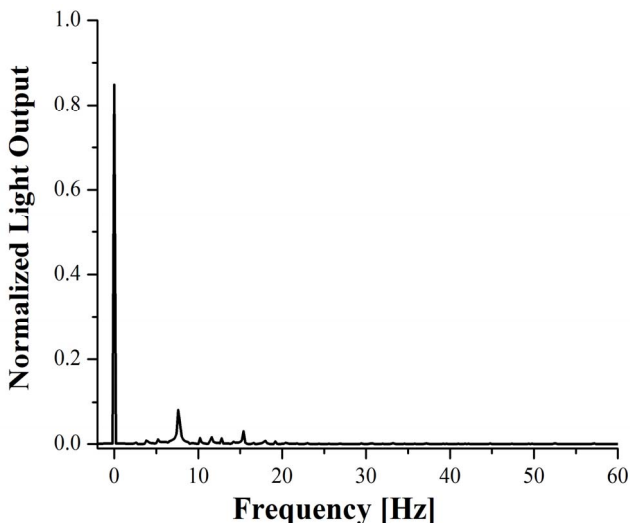


Fig. 9. Light spectrum obtained from FFT decomposition in the case of light emitted by a PC monitor.

of the video terminal but the interest region can be chosen by selecting the interest sections of the matrices corresponding to the recorded images.

V. CONCLUSIONS

The proposed method can easily provide information concerning the light flicker induced by conventional light sources but it can be also used for detecting the light flicker that may be generated by some computer software (games especially) or TV programs. One of the main advantages of the method in this case is that the matrix corresponding to each image can be divided in smaller sections (e.g. quarters) for detecting the light flicker in different regions of the screen.

The method was implemented and tested for three different conventional light sources – magnetically ballasted fluorescent lamp, SSL and incandescent lamps – as well as for the cases of a PC monitor displaying a flicker generating program.

The waveforms corresponding to the light intensity variation in time for the conventional light sources are similar to the ones presented in literature but the values obtained for the flicker index and percentage flicker might present high errors because a light intensity calibration using a standardized light source was not performed.

Future work will be dedicated to more accurate measurements involving the system calibration as well as comparing the results with the ones provided by specialized sensors.

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